NAVYA SYAL

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LinkedIn |Digital Portfolio | Github |

➢ OBJECTIVE

I'm a Pre - Final Year Cybersecurity undergraduate with knowledge on **Network Security**, **OWASP**, **Cyber Forensics, Azure Cloud Security** complemented by hands-on experience in **Frontend Development using React and Vite**. I aim to build a career in **SOC Operations**, **Threat Detection**, **Log Analysis**, and **Incident Response**. Currently preparing for CompTIA Security+ Certification and seeking a 1-year Cybersecurity or SOC Internship.

➢ PROJECTS

1. ZERO TRUST ARCHITECTURE WITH HYBRID CLOUD (ONGOING PROJECT)
    - Developing a framework based on Zero Trust principles for a hybrid cloud environment.
    - Implemented RBAC to allow permissions based on user roles.
    - Integrates authentication and authorization to prevent unauthorized access

2. FORENSIC-AI: AN INTELLIGENT DEEPFAKE VIDEO DETECTION FOR DIGITAL EVIDENCES (ONGOING PROJECT)
    - Evaluating cross dataset generalization to test model performance on unseen datasets.
    - Building a Deep Learning Model based on Vision Transformer (ViT) to detect deepfake videos
    - Aiming to improve digital forensic authentication of video evidence

3. CAREER COUNSELING MANAGEMENT SYSTEM
    - Developed a database management system to provide career guidance.
    - Implements data storage, retrieval, and management using database technologies.

4. EXPLORING THE MAZE: A COMPARATIVE STUDY OF PATH FINDING ALGORITHMS FOR PAC-MAN GAME
    - Conducted a comparative analysis of search algorithms using path findings.
    - Implemented DFS, BFS, UCS and A* algorithms in PAC-MAN game.
    - Evaluated algorithm-based performance across different maze sizes and complexities.

5. QUANTUM ANALYSIS OF AES USING SHANNON'S THEORY
    - Performed a cryptographic research analysis of AES encryption variants (AES-128, AES-192, AES-256)
    - Evaluated security using Shannon's Information Theory concepts such as entropy and redundancy.

6. ATDAM: ADAPTIVE THREAT DETECTION AND AUTOMATED MITIGATION FRAMEWORK (RESEARCH STUDY)
    - Proposed a cybersecurity framework for adaptive threat detection and automated mitigation.
    - Combines signature-based and anomaly-based techniques.
    - Designed to identify threats faster and reduce attacker dwell time.

➢ CERTIFICATIONS

1. CYBERSECURITY ANALYST JOB SIMULATION – TATA CONSULTANCY SERVICES (FORAGE)
   - Worked on IAM fundamentals and strategy assessment.
   - Explored IAM integration and security solution implementation.

2. CYBERSECURITY JOB SIMULATION – MASTERCARD (FORAGE)
   - Completed simulation as a part of Mastercard's Security Awareness Team
   - Identified and analysed phishing threats.

3. INTRODUCTION TO CYBERSECURITY – IBM SKILLSBUILD
   - Learned core cybersecurity concepts, common threats and defensive practices.

➢ EDUCATION

BTech in Cyber Security, NIIT University **2023 to 2027**

Relevant Modules: Cyber Security, Operating Systems, Computer Network and Data Communications, Secure Programming and Application Security, Network Security, Cyber Forensics, Cryptography, Vulnerability Assessment and Penetration Testing, Identity and Access Management, Database Security Monitoring and Auditing.

XII, Somerville School, Vasundhara Enclave (CBSE) **2022 - 2023**

X, Somerville School, Vasundhara Enclave (CBSE) **2020 – 2021**

➢ SKILLS

- Cloud Security
- SQL, MySQL
- HTML
- Linux
- MERN Stack
- Forensic Analysis
- Vulnerability Scanning
- Cybersecurity Fundamentals
- Zero Trust Architecture
- Network Security
- Identity and Access Management (IAM)
- Git & GitHub
- Jenkins (CI/CD)
- Python

➢ POSITIONS OF RESPONSIBILITY

- **President, Agraga - The Management Society of NIIT University**

  Coordinated and managed university events, overseeing logistics, scheduling, and participant management while collaborating with faculty and student teams.

- **Member, Student Affairs Committee (SAC) at NIIT University**

  Supported student well-being and helped address student needs as a member of the Student Affairs Committee (SAC)